

Exam Date & Time: 17-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: ICT 3122/IT_3122 PRINCIPLES OF OPERATING SYSTEMS

(SPL: INFORMATION TECHNOLOGY)

PRINCIPLES OF OPERATING SYSTEMS [ICT 3122]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Two processes are trying to enter their critical sections simultaneously using Dekker's Version 3 algorithm. Apply the algorithm's logic to explain why a deadlock can occur in this situation. Write the supporting pseudocode to justify your answer. (5)
- 1B) Your college must launch a student coding portal within one week. Usage will be low most days but surge during exams. The team wants to avoid buying servers, keep operations minimal, and pay only for what is used.
i) Choose an environment category for deployment (e.g., on-premises data center, hosted/on-demand provider, or a mix). (3)
ii) Choose a service option: a fully managed application platform, self-managed virtual machines, or a ready-made application subscription.
iii) Identify the network type that can be used by the college. Justify your choices.
- 1C) In a multi-programmed operating system, several processes frequently request access to shared resources such as printers, disk drives, and memory. Occasionally, some processes get stuck indefinitely because each process is holding one resource and waiting for another that is already held by another process. This situation leads to a deadlock, where none of the involved process can proceed. Apply the three deadlock-handling methods to the given scenario for handling deadlocks. (2)
- 2A) Consider the following reference string for a demand-paged system: Reference String : 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6.
i) Illustrate Belady's Anomaly using the given reference string and show step-by-step page replacement, when the number of free frames in the memory are three and four. (5)
ii) What are the problems associated with LRU page replacement algorithm? Justify with an example.
iii) Analyze the effect of increasing the degree of multiprogramming beyond a certain limit in a demand-paged system?
- 2B) Apply your understanding of Linux system structure to explain how a user program reads a file from disk, detailing the roles of system libraries, the kernel, and system utilities in this operation. (3)
- 2C) A video rendering process is executing on the CPU. When its time quantum expires, the CPU scheduler preempts it and moves another process to the CPU. Identify the new state of the video rendering process after preemption and illustrate this transition using a process state diagram. (2)
- 3A) A system has four processes (P_1, P_2, P_3, P_4) and four resources (R_1, R_2, R_3, R_4). Each process currently holds and requests certain resources as shown in Table Q. 3A. (5)

Table Q. 3A

Process	Currently Holding	Currently Requesting
P ₁	R ₁	R ₂
P ₂	R ₂	R ₃
P ₃	R ₃	R ₁
P ₄	R ₄	R ₂

Using this information, analyze the following situations:

- Construct a Wait-For Graph (WFG) for the given system and identify all the deadlocked processes.
- Suggest the minimal resource or edge change that would break the deadlock without terminating all processes. Specify which process or resource should be altered and why.

- 3B) Consider the exponential average formula used to predict the length of the next CPU burst.
- What are the implications of assigning the following values to the parameters used by the algorithm?
 - $\alpha = 0$ and $\tau = 100$ milliseconds
 - $\alpha = 0.99$ and $\tau = 10$ milliseconds
 - Given the following sequence of CPU-burst times, use the exponential average formula to predict the length of the next CPU burst. Assume $\tau_0 = 10$ ms and $\alpha = 0.5$.
CPU burst (t_i) : 8 6 8 6 15 15 15
- 3C) A system has $m=200$ physical frames, the OS reserves 8 frames; the remaining are available to 5 user processes with the following sizes: $s_1=65$, $s_2=140$, $s_3=85$, $s_4=210$, $s_5=100$
- Compute the frame allocation for each process using proportional allocation.
 - How equal allocation may impact system performance in this case.
- 4A) A process consists of three memory segments: a Code segment of 1000 bytes, a Data segment of 500 bytes, and a Stack segment of 200 bytes. The available memory blocks in the system are of the following sizes: Block 1 has 1200 bytes, Block 2 has 800 bytes, and Block 3 has 500 bytes.
- Apply segmentation allocation to assign each segment to a suitable memory block.
 - Compute the external fragmentation resulting from this allocation.
 - If the system uses first-fit allocation strategy, explain how the segments would be placed.
 - If the system uses best-fit allocation strategy, explain how the segments would be placed.
 - Suppose a new segment of 600 bytes needs to be added. Can it be allocated? If yes, to which block? If not, explain why.
- 4B) Apply the C-SCAN, LOOK and SSTF disk scheduling algorithm to the following sequence of disk requests: 182, 270, 43, 140, 24, 16, 290. Assume the disk is moving towards higher cylinders. The disk head starts at position 150. Show the schematic representation and calculate the total head movement. Analyse which algorithm is better for uniformly distributed requests.
- 4C) Identify real-world applications that typically use sequential access or random access. Justify your answer.
- Media player
 - Database management system
 - Log file analyzer
 - Compiler intermediate file

- 5A) Consider a computer system that uses demand paging. The memory access time is 100 nanoseconds, and the page fault service time is 8 milliseconds. In this system, 80 percent of the time the replaced pages are not modified and take 6 milliseconds to service, while the remaining 20 percent are modified and require 10 milliseconds to service. The page-fault rate (p) is 0.002, corresponding to 0.2 percent. List and briefly describe the steps involved in handling a page fault in a demand-paging system. (5)
 Compute the average page fault service time considering both modified and unmodified pages.
 Using your result from (b), calculate the Effective Access Time (EAT). Then, evaluate whether this system's performance is acceptable if the target $EAT \leq 200$ nanoseconds.
- 5B) Given the following tasks as shown in Table Q. 5B:
- Table Q.5B**
- | Task | Execution Time (ms) | Period (ms) |
|------|---------------------|-------------|
| T1 | 2 | 4 |
| T2 | 4 | 8 |
- (3)
- i) Compute the CPU utilization for both the Rate-Monotonic Scheduling (RMS) and Earliest-Deadline-First (EDF) algorithms.
 ii) Identify the scheduling approach that provides a better real-time guarantee for this task set.
- 5C) A factory automation system must control robotic arms with strict timing, a temperature-monitoring system that can occasionally delay updates, and a reporting module that logs data once per minute. Classify each module as hard, firm, or soft real-time, and justify your classification based on timing and scheduling requirements. (2)

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Event Name	Subject
058-END SEMSTER-ODD-2025	IT_3122:PRINCIPLES OF OPERATING SYSTEM

#	Course Outcome	Description	Marks
1	1	CO 1	6
2	2	CO 2	27
3	3	CO 3	12
4	4	CO 4	5
Total			50

* NA - CO Not Selected

Event Name	Subject
058-END SEMSTER-ODD-2025	IT_3122:PRINCIPLES OF OPERATING SYSTEM

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	13	40(80%)

4	Analysis	2	10(20%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)